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Geometric-Robust Contrastive Learning Framework with Layout Topology Representation and Self-Calibration Voting for Hotspot Detection

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***Abstract*—Recently, machine learning-based techniques have been increasingly employed for layout hotspot detection. However, existing supervised-based methods often struggle to learn a reliable decision boundary across the entire dataset and ignore the geometric properties and topology of the polygons. Moreover, robustness to geometric transformations of layout patterns has received limited attention in hotspot detection, potentially affecting model generalization. To address these challenges, this article introduces GRC-HD, a novel contrastive learning framework that aligns image features with layout topology representations while integrating geometric-robust feature learning. The framework improves the ability to distinguish between hotspots and non-hotspots by similarity computations instead of a single decision boundary. To effectively incorporate geometric information into the model training process, we propose Layout2Seq, which encodes polygon shapes as vector sequences and feeds them into GRC-HD. Furthermore, to better represent topological information, we develop an absolute position embedding, replacing the standard position encoders used in Transformer architectures. To improve the robustness of hotspot detector, we design a configurable geometric transformation library that generates multiple manufacturing-relevant views. The library is leveraged through an auxiliary Siamese network during training for geometric-robust representation learning, and through a self-calibration voting strategy at inference, where transformed instances act as independent voters to enhance decision reliability. Extensive experiments on various benchmarks demonstrate that GRC-HD outperforms current state-of-the-art methods.**

***Index Terms*—Design for manufacturability, contrastive learning, hotspot detection, layout representation, deep learning.**

I. Introduction

With the rapid advancement of semiconductor technology, the continuous scaling of transistor feature sizes and the increasing complexity of integrated circuits pose significant challenges for Design for Manufacturability (DFM) [1]. Hotspot detection is a critical DFM step for identifying potential defect patterns early in the design flow. Traditional lithography simulation technique [2] can accurately detects hotspot patterns, but it is computationally expensive and is often impractical for large-scale deployment in design flows [3], [4]. As an alternative, pattern-matching approaches detect hotspots by retrieving or clustering similar patterns. Although effective when hotspots are well represented in predefined pattern or layout libraries, these methods generalize poorly and often fail to identify previously unseen hotspot patterns [5], [6].

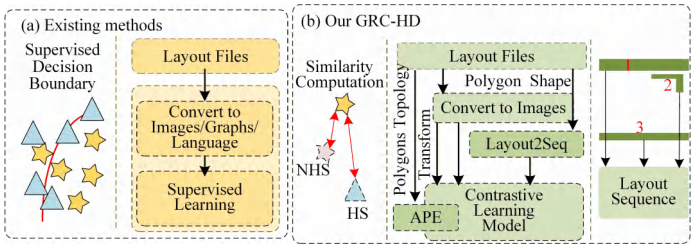

Fig. 1. Illustration of the hotspot detection flow. (a) The pipeline of existing supervised learning-based methods. (b) The pipeline of our proposed GRC-HD.

Recently, machine learning-based methods have been applied in DFM [7]–[9] and achieved impressive progress in hotspot detection [10]–[12]. As shown in Fig. 1(a), existing methods predominantly rely on supervised learning to classify HotSpot (HS) and Non-HotSpot (NHS). However, these supervised learning approaches often suffer from a high false-alarm rate due to the inherent imbalance in hotspot datasets and the limited inter-class variance. Specifically, training on the full dataset with a limited number of hotspot samples can lead to suboptimal generalization [10], [13]. Moreover, subtle geometric differences between hotspot and non-hotspot patterns make it difficult to learn a reliable decision boundary. As shown in Fig. 1(a), beyond the limitations of supervised learning, existing methods also exhibit constraints in effectively modeling layout topology. Most convolutional neural network (CNN)–based hotspot detection methods are constrained by the receptive field, which limits their ability to capture global topological information from layout images [14]. To address this, some methods represent layouts as graphs using polygon-based structures and apply graph neural networks (GNNs) for hotspot classification [15], [16]. Nevertheless, these methods decompose polygons into multiple rectangular nodes, increasing computational complexity and leading to the potential loss of critical shape details of the

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original polygons. The squish pattern representation describes layout clips through a topological matrix combined with two vectors [8], [9], [17]. However, this representation exhibits limited flexibility when integrated with existing deep neural network architectures [18]. More recently, a sequence-based pattern representation [18] was proposed to model layout, enabling the learning of shape and spatial attributes and facilitating the identification of hotspots using Transformer-based models [12]. Nevertheless, converting continuous coordinates and edge attributes into discrete token embeddings introduces a representation mismatch, and the method does not explicitly encode polygon positions. To the best of our knowledge, the shapes of individual polygons (e.g., height, width, orientation) and the topological relationships among polygons (e.g., edge to edge, side to side) are both critical for identifying hotspots [9]. Thus, the aforementioned supervised learning methods and layout modeling strategies struggle to construct robust statistical models for hotspot detection.

During physical design, instances are often mirrored or rotated to improve placement density and routability. In layout analysis and clustering, layout snippets related by these geometric transformations are typically regarded as equivalent for matching, which significantly reduces the number of unique samples that must be analyzed [19]. For the machine learning-based hotspot detector, conventional neural network architectures, particularly CNNs, lack rotation-invariant feature extraction capabilities. To address this limitation, existing methods commonly employ data augmentation, expanding training datasets by applying geometric transformations to the original training samples [13], [20]–[23]. However, such augmentation increases computational cost and training time. Moreover, models trained through augmentation do not inherently learn robustness but instead rely on the memorization of augmented samples. In other words, the performance will degrade when encountering unseen transformations not explicitly included in training. Several methods [19], [24] leverage self-contrastive learning to obtain transformation-aware layout features. However, these methods generally require additional fine-tuning for hotspot detection, as they do not provide end-to-end geometric-robust hotspot detection capability.

Considering the challenges discussed above, this article proposes a novel geometric-robust contrastive learning framework, GRC-HD, designed to align image features with layout topology representations effectively. As illustrated in Fig. 1(b), the proposed framework leverages contrastive learning to obtain a unified representation by integrating visual features with geometric polygon sequences. Moreover, we introduce a Siamese network to align representations across configurable geometric layout transformations, thereby extracting geometric-robust layout features. To encode the geometric structure of polygons comprehensively, GRC-HD introduces a novel module, Layout2Seq, which transforms the geometric attributes of individual polygons into embedding vectors. Additionally, we propose an Absolute Position Embedding (APE) method to explicitly incorporate the centroid coordinates to capture topology information critical for accurate layout hotspot detection. In the inference phase, the GRC-HD framework discriminates hotspots from non-hotspots based on feature similarity computations rather than relying on a single decision boundary. To further enhance prediction reliability, we devise a self-calibration voting mechanism that aggregates results from multiple layout transformations. The main contributions of this article are summarized as follows:

- We propose a novel contrastive learning framework for hotspot detection, that leverages similarity computations, reducing reliance on a single global decision boundary over the entire dataset.
- We present Layout2Seq, which combines moment-based shape statistics with geometric attributes to convert layout polygons into a sequence representation, facilitating the extraction of geometric features. This representation is also applicable to other machine learning-based DFM tasks.
- We develop the Absolute Position Embedding (APE) method for Transformer-based sequence models to efficiently encode the topological relationships of multiple polygons based on their position context within the layout.
- We introduce a geometric-robust learning framework that integrates Siamese contrastive learning to encourage feature consistency under configurable geometric transformations. In addition, we employ self-calibration voting via ensemble inference to improve prediction reliability.
- Experimental results demonstrate that GRC-HD significantly improves the discrimination of hotspots through contrastive methods, achieving a maximum accuracy improvement of 6.3% and a reduction in the false alarm rate by up to 25.78%. Furthermore, GRC-HD can be effectively fine-tuned with linear layers, yielding remarkable performance in hotspot detection.

II. Preliminaries

The lithographic process employs a photomask to transfer precisely defined layout patterns onto semiconductor wafers. Sensitive patterns are particularly susceptible to yield reduction caused by variations in the manufacturing process. These patterns are defined as hotspots within the layout [11]. In this article, hotspot detection is considered as a binary classification task performed on layout clips. Different from [10], [16], [24], we consider layout clips in which hotspot locations are random and may include multiple hotspots, rather than just a single centered marker. It will test the model's ability to learn non-fixed hotspot patterns making it more relevant to real-world scenarios. In addition, we apply geometric transformations to the evaluation clips during testing to ensure robustness. The following definitions and metrics are used to evaluate the performance of a hotspot detector.

Definition 1 (Accuracy). The ratio of correctly identified hotspots to the total number of ground truth hotspots.

Definition 2 (False Alarm). The ratio of non-hotspot clips that are detected as hotspots by the classifier.

To match the evaluation metrics above, we formulate the hotspot detection problem as follows.

Problem 1 (Hotspot Detection). Given a collection of layout clips containing hotspot and non-hotspot patterns, our goal

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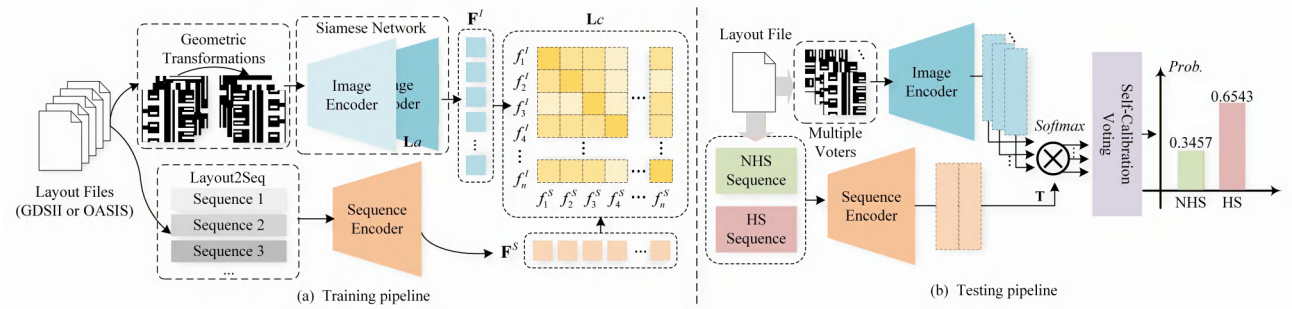

Fig. 2. The overview of GRC-HD. (a) The training process of the contrastive learning for the hotspot detection. (b) The simple testing process of the proposed framework.

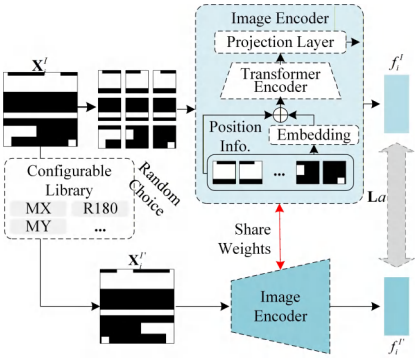

Fig. 3. Illustration of the Siamese network in training process.

is to learn a feature vector by constructing both image and sequence representations. Based on this vector we can use contrastive learning to distinguish hotspots or fine-tune a classifier for detecting hotspots.

III. Methodology

A. Overview

As shown in Fig. 2(a), the proposed training pipeline first generates the image-based representations of the layouts, consistent with prior work [3], [14], [19]. We then introduce Layout2Seq, which serializes polygons into a sequence representation by encoding each polygon using moment-based shape statistics with basic geometric attributes. A Siamese network and a sequence encoder are then employed to extract geometrically robust image and sequence features, respectively. Then, these features are aligned using a contrastive loss, producing a unified embedding that integrates the image and sequence representations. As shown in Fig. 2(b), during inference we generate two candidate sequences (hotspot and non-hotspot) for each test layout using Layout2Seq. In addition, we apply a set of transformations to obtain multiple voters from the input image. We then perform a self-calibration voting via ensemble inference that aggregates the classification results from these voters, thereby improving the reliability and accuracy of hotspot discrimination.

B. Geometric-Robust Contrastive Learning Framework

As shown in Fig. 2, our proposed framework consists mainly of two key components, specifically image encoder $\mathcal{I}(\cdot)$ and sequence encoder $\mathcal{S}(\cdot)$. Motivated by [25], our objective is to jointly train $\mathcal{I}(\cdot)$ and $\mathcal{S}(\cdot)$ using pairs of binary layout images and corresponding sequence representations, ensuring that the learned image embedding $\mathbf{F}^I = [f_1^I, f_2^I ..., f_N^I] \in \mathbb{R}^{N\times D}$ are effectively aligned with the sequence embedding $\mathbf{F}^S \in \mathbb{R}^{N\times D}$, where N denotes the batch size and D represents the embedding dimension. Moreover, the well-trained image encoder $\mathcal{I}(\cdot)$ is capable of extracting geometric-robust features from input images.

We first train the image encoder using a Siamese architecture to improve robustness and generalization under diverse geometric transformations. As shown in Fig. 3, given an input layout image $\mathbf{X}_i^I$, we randomly sample a transformation from a predefined library of geometric operations and apply it to generate the augmented image $\mathbf{X}_i^{I'}$. Considering that pattern transfer under different process conditions and illumination settings may deviate from strict geometric equivalence, we adopt a configurable library of geometric operations to generate an enhanced image representation. In this work, the candidate transformations include rotations by $90°$, $180°$, and $270°$, denoted as R90, R180, and R270, as well as mirror operations. Appropriate combinations of these transformations can be selected to accommodate design- and process-dependent characteristics. For example, for advanced technology nodes we may apply a subset consisting of R180 together with X mirror (MX) and Y mirror (MY) operations. Subsequently, both the original layout image and its transformed counterpart are fed into the Siamese image encoder with shared weights, producing two latent representations. We employ a Vision Transformer (ViT) as the image encoder because it preserves spatial relationships and captures long-range contextual dependencies. This capability is essential for achieving accurate and meaningful feature representations in layout [14]. The detailed architecture of ViT is shown in Fig 3. The features extracted by the image encoder are subsequently mapped into an embedding space via a projection layer. The final embeddings for both the original and augmented images are calculated by:

$$f_i^I, f_i^{I'} = \mathrm{Proj}(\mathcal{I}(\mathbf{X}_i^I)), \mathrm{Proj}(\mathcal{I}'(\mathbf{X}_i^{I'})), \tag{1}$$

where f_i^I and $f_i^{I'}$ denote the embeddings corresponding to $\mathbf{X}_i^I$ and $\mathbf{X}_i^{I'}$, respectively. The function $\mathrm{Proj}(\cdot)$ represents a projection layer. The encoders $\mathcal{I}$ and $\mathcal{I}'$ share identical network

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Algorithm 1 Training process for our proposed GRC-HD

Require: Training pair set of layout representations $\{\mathbf{X}_i^I, \mathbf{X}_i^S\}_{i=0}^N$, where $\mathbf{X}_i^S$ is from Algorithm 2, a predefined library of geometric operations $\{op(\cdot)\}$ and the epoch E.

Parameters: An image encoder $\mathcal{I}$, a sequence model $\mathcal{S}$.

Initialize $\mathcal{I}$, $\mathcal{S}$ randomly.
for $e = 0$ to E **do**
  for $i = 1$ to N **do**
    $op(\cdot) \leftarrow$ Random choice in $\{op(\cdot)\}$;
    $\mathbf{X}_i^{I'} \leftarrow op(\mathbf{X}_i^I)$;
    $(f_i^I, f_i^{I'}) \leftarrow$ Compute the embedding by $\mathcal{I}$ and Equation (1)
    $L_a \leftarrow$ Compute the alignment loss by $(f_i^I, f_i^{I'})$ and Equation (2);
    $f_i^S \leftarrow$ Compute the embedding by the $\mathcal{S}$ and Equation (3);
    $L_{I\to S}, L_{S\to I} \leftarrow$ Compute the image-to-sequence loss and sequence-to-image loss by $(f_i^I, f_i^{I'})$ and Equation (4) and Equation (5);
    $L_c \leftarrow$ Compute the contrastive loss by the Equation (7);
    Optimize $\mathcal{I}, \mathcal{S}$ by L_a, L_c and Equation (8);
  end for
end for

weights. In the training process, we use an alignment loss based on cosine similarity to encourage geometric robustness:

$$L_a = \frac{1}{N}\sum_{i=1}^{N} 1 - \frac{\langle f_i^I, \left(f_i^{I'}\right)^{\mathrm{T}}\rangle}{\left\|f_i^I\right\| \times \left\|f_i^{I'}\right\|}, \tag{2}$$

where $\langle\cdot,\cdot\rangle$ denotes the vector inner product, $\|\cdot\|$ denotes the vector norm and $(\cdot)^{\mathrm{T}}$ indicates the transpose operation.

In the sequence branch, the sequence encoder $\mathcal{S}(\cdot)$ is implemented as a Transformer encoder [26] with absolute positional embeddings. Given the Layout2Seq output, $\mathcal{S}(\cdot)$ produces sequence features $\mathbf{F}^S$. Specifically, for an input sequence $\mathbf{X}_i^S$, the embedding f_i^S is computed as:

$$f_i^S = \mathrm{Proj}(\mathcal{S}(\mathbf{X}_i^S)), \tag{3}$$

To align the visual embeddings $\mathbf{F}^I$ with the sequence embeddings $\mathbf{F}^S$, we define the image-to-sequence contrastive loss over paired samples $(\mathbf{X}_i^I, \mathbf{X}_i^S)$ as:

$$L_{I\to S} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\mathrm{sim}(f_i^I, f_i^S)/\tau)}{\sum_{j=1}^{N}\exp(\mathrm{sim}(f_i^I, f_j^S)/\tau)}, \tag{4}$$

and the reversed sequence-to-image contrastive loss $L_{S\to I}$ are obtained by:

$$L_{S\to I} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\mathrm{sim}(f_i^S, f_i^I)/\tau)}{\sum_{j=1}^{N}\exp(\mathrm{sim}(f_i^S, f_j^I)/\tau)}, \tag{5}$$

where the τ is the temperature parameter and $\mathrm{sim}(\cdot)$ denotes the similarity function which is calculated as:

$$\mathrm{sim}(f_i^I, f_i^S) = \frac{\langle f_i^I, \left(f_i^S\right)^{\mathrm{T}}\rangle}{\left\|f_i^I\right\| \times \left\|f_i^S\right\|}. \tag{6}$$

Algorithm 2 Layout2Seq

Require: a set of polygons for the i-th layout clip $\{P^k\}_{k=0}^l$, the polygon number of the layout clip l, the sequence context length L, the label of the i-th layout $\hat{Y}$.

Ensure: the sequence for the i-th layout clip $\mathbf{X}_i^S$.

$\mathbf{X}_i^S \leftarrow ()$, Initialize empty sequences;
for $k = 1$ to l **do**
  $\mathbf{V}_i^k \leftarrow ()$, Initialize empty polygon vector;
  $I^k \leftarrow$ Rendering a binary image by polygon P^k;
  $\tilde{\mathbf{M}}_{pq}^k \leftarrow$Compute by Equation(9)(10)(11) and I^k;
  $\mathbf{Hu}_i^k \leftarrow$ Compute by Equation(12);
  $v_1 \sim v_7 \leftarrow \tilde{\mathbf{M}}_{20}^k, \tilde{\mathbf{M}}_{11}^k, \tilde{\mathbf{M}}_{02}^k, \tilde{\mathbf{M}}_{30}^k, \tilde{\mathbf{M}}_{21}^k, \tilde{\mathbf{M}}_{12}^k, \tilde{\mathbf{M}}_{03}^k$;
  $v_8 \sim v_{14} \leftarrow \mathbf{Hu}_1, \mathbf{Hu}_2, \mathbf{Hu}_3, \mathbf{Hu}_4, \mathbf{Hu}_5, \mathbf{Hu}_6, \mathbf{Hu}_7$;
  $\mathbf{V}_i^k \leftarrow \{v_1 \sim v_{14}\}$;
  $a \leftarrow \mathrm{Area}(P^k)$;
  $p \leftarrow \mathrm{Perimeter}(P^k)$;
  $w_{bbox}, h_{bbox} \leftarrow \mathrm{getBoundingRect}(P^k)$;
  $v_{15} \leftarrow p/a$;
  $v_{16} \leftarrow a/(w_{bbox} \times h_{bbox})$;
  $v_{17} \leftarrow \mathrm{isContourConvex}(P^k)$;
  $\mathbf{V}_i^k \leftarrow \{v_{15}, v_{16}, v_{17}\}$;
  $\mathbf{X}_i^S \leftarrow \mathbf{V}_i^k$;
end for
$l_{cur} = len(\mathbf{X}_i^S)$;
for $l = l_{cur}$ to L **do**
  if $\hat{Y} = HS$ **then**
    $\mathbf{X}_i^S \leftarrow \{1\}^{17}$;
  else if $\hat{Y} = NHS$ **then**
    $\mathbf{X}_i^S \leftarrow \{-1\}^{17}$;
  end if
end for
return $\mathbf{X}_i^S$.

The contrastive loss L_c in Fig 2(a) is calculated as:

$$L_c = \frac{L_{I\to S} + L_{S\to I}}{2}. \tag{7}$$

This loss function is designed to optimize the model by increasing the similarity between matched pairs of representations, (f_i^I, f_i^S), while simultaneously decreasing the similarity between mismatched pairs, (f_i^I, f_j^S), for $i \neq j$. Additionally, the contrastive learning strategy employed by GRC-HD encourages compact embeddings for semantically similar samples, while ensuring clear separation between the embeddings of dissimilar samples. Finally, the total loss L_{total} for the proposed GRC-HD is calculated as:

$$L_{total} = L_a + L_c, \tag{8}$$

and the training process for geometric-robust contrastive learning is presented in Algorithm 1.

C. Layout to Sequence

As shown in Fig. 4, several layout representations have been adopted in deep learning methods. One of the most straightforward methods is to convert a layout directly into a binary image. To improve efficiency, a compressed encoding

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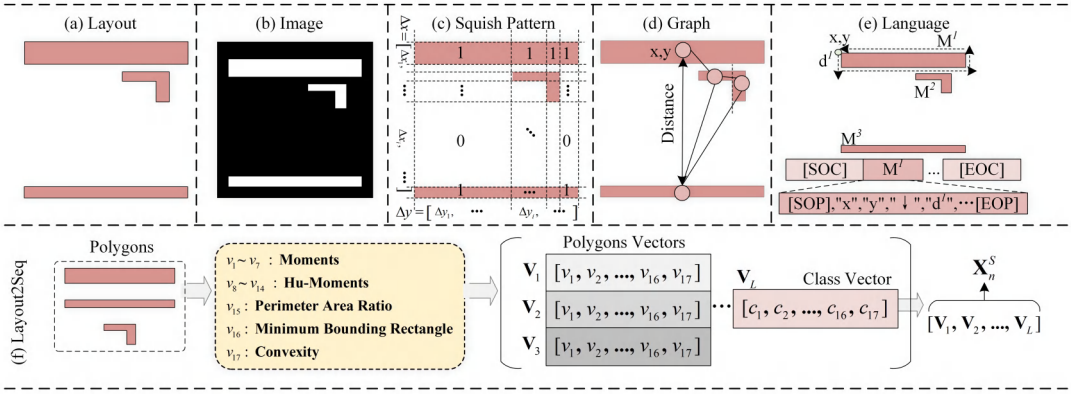

Fig. 4. Comparison of various layout representations: (a) Original layout. (b) Binary image representation of the layout. (c) Topological matrix accompanied by positional vector representation. (d) Graph-based representation. (e) Natural language-based representation. (f) The proposed Layout2Seq representation, transforming the layout into a geometric sequence.

method known as the squish pattern has also been proposed. However, both binary image and squish pattern representations typically rely on CNNs, which are inherently constrained by their local receptive fields. Graph-based representations offer a more flexible modeling framework but require an initial polygon decomposition step. This preprocessing is often complex and computationally intensive. More recently, natural language-based representations have been developed, which require the tokenization of numerical polygon attributes. However, the geometric properties of polygons differ from the discrete tokens found in natural language corpora, which may constrain the effectiveness of language-based representations.

To effectively capture geometric information from polygons, we introduce Layout2Seq, which converts each layout clip into a fixed-length sequence representation of polygons. Each sequence comprises moment-based descriptors and basic geometric attributes, enabling more effective extraction and analysis of geometric features. As shown in Fig. 4(f), we extract polygon coordinates from the original layout files and systematically parse each polygon using a vectorized polygon method. Image moments are widely used in computer vision and image processing to characterize object shapes [27]. Therefore, for each polygon P^k, where k is the index of the polygon, we generate a binary image I^k using the polygon's coordinates. Subsequently, the geometric moment $\mathbf{M}^k$ of the I^k is computed by:

$$\mathbf{M}_{pq}^k = \sum_{x=0}^{W} \sum_{y=0}^{H} x^p \cdot y^q \cdot I^k(x, y), \tag{9}$$

where W and H represent the width and height of I^k, p and q denote the order of the moment. We further compute the central moments $\mathbf{C^k}$ as:

$$\mathbf{C}_{pq}^k = \sum_{x=0}^{W} \sum_{y=0}^{H} (x - \frac{\mathbf{M}_{10}^k}{\mathbf{M}_{00}^k})^p \cdot (y - \frac{\mathbf{M}_{01}^k}{\mathbf{M}_{00}^k})^q \cdot I^k(x, y), \tag{10}$$

where $\mathbf{M}_{00}$, $\mathbf{M}_{10}$ and $\mathbf{M}_{01}$ can be used to compute the x-coordinate and y-coordinate of the centroid. For scale-invariant moments which are useful for comparing shapes regardless of their size, we compute the normalized moments $\tilde{\mathbf{M}}^k$ as follows:

$$\tilde{\mathbf{M}}_{pq}^k = \frac{\mathbf{C}_{pq}^k}{(\mathbf{M}_{00}^k)^{(p+q)/2+1}}. \tag{11}$$

The $\{\tilde{\mathbf{M}}_{20}^k, \tilde{\mathbf{M}}_{11}^k, \tilde{\mathbf{M}}_{02}^k, \tilde{\mathbf{M}}_{30}^k, \tilde{\mathbf{M}}_{21}^k, \tilde{\mathbf{M}}_{12}^k, \tilde{\mathbf{M}}_{03}^k\}$ are incorporated into the polygon vector $\mathbf{V}^k$ as components $\{v_1 \sim v_7\}$. In addition, we further utilize Hu's seven moments [28] to capture shape invariance under common geometric transformations. The Hu moments can be calculated as:

$$\begin{aligned}
\mathbf{Hu}_1^k &= \tilde{\mathbf{M}}_{20}^k + \tilde{\mathbf{M}}_{02}^k, \\
\mathbf{Hu}_2^k &= (\tilde{\mathbf{M}}_{20}^k - \tilde{\mathbf{M}}_{02}^k)^2 + 4(\tilde{\mathbf{M}}_{11}^k)^2, \\
\mathbf{Hu}_3^k &= (\tilde{\mathbf{M}}_{30}^k - 3\tilde{\mathbf{M}}_{12}^k)^2 + (3\tilde{\mathbf{M}}_{21}^k - \tilde{\mathbf{M}}_{03}^k)^2, \\
\mathbf{Hu}_4^k &= (\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)^2 + (\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k)^2, \\
\mathbf{Hu}_5^k &= (\tilde{\mathbf{M}}_{30}^k - 3\tilde{\mathbf{M}}_{12}^k)(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)\Big[(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)^2 \\
&\quad - 3(\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k)^2\Big] + (3\tilde{\mathbf{M}}_{21}^k - \tilde{\mathbf{M}}_{03}^k)(\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k) \\
&\quad \Big[3(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)^2 - (\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k)^2\Big], \\
\mathbf{Hu}_6^k &= (\tilde{\mathbf{M}}_{20}^k - \tilde{\mathbf{M}}_{02}^k)\Big[(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)^2 - (\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k)^2\Big] \\
&\quad + 4\tilde{\mathbf{M}}_{11}^k(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)(\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k), \\
\mathbf{Hu}_7^k &= (3\tilde{\mathbf{M}}_{21}^k - \tilde{\mathbf{M}}_{03}^k)(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)\Big[(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)^2 \\
&\quad - 3(\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k)^2\Big] - (\tilde{\mathbf{M}}_{30}^k - 3\tilde{\mathbf{M}}_{12}^k)(\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k) \\
&\quad \Big[3(\tilde{\mathbf{M}}_{30}^k + \tilde{\mathbf{M}}_{12}^k)^2 - (\tilde{\mathbf{M}}_{21}^k + \tilde{\mathbf{M}}_{03}^k)^2\Big].
\end{aligned} \tag{12}$$

The $\{\mathbf{Hu}_1^k, \mathbf{Hu}_2^k, \mathbf{Hu}_3^k, \mathbf{Hu}_4^k, \mathbf{Hu}_5^k, \mathbf{Hu}_6^k, \mathbf{Hu}_7^k\}$ are incorporated into the polygon vector $\mathbf{V}^k$ as components $\{v_8 \sim v_14\}$. Additionally, we have incorporated area-perimeter ratio v_{15}, minimum bounding rectangle v_{16}, and convexity v_{17} to more accurately characterize the geometric properties of polygons. Consequently, for each polygon, we derive a comprehensive vector of geometric information, denoted as $\mathbf{V}_i^k$, which includes a total of 17 values.

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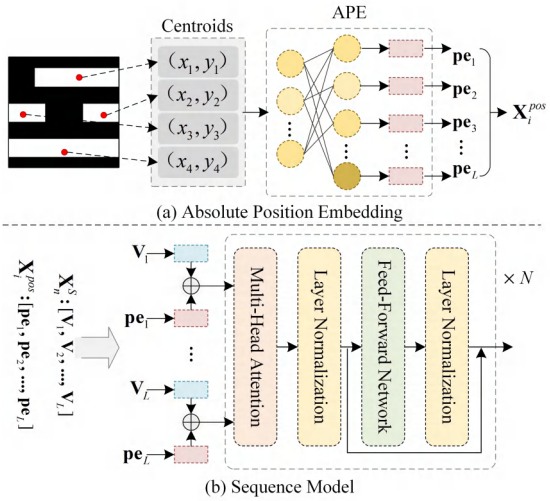

Fig. 5. Illustration of absolute position embedding and sequence model.

Algorithm 3 The detail of sequence model $\mathcal{S}(\cdot)$ with APE

Require: The sequence representation of layout $\mathbf{X}_i^S$, the polygon number of the layout clip l, the sequence context length L.

Ensure: the output f_i^S of $\mathcal{S}(\cdot)$.

for $k = 1$ to l **do**
  $x_k, y_k \leftarrow$ Compute the centroid coordinates by Equation(13);
  $\hat{x}_k, \hat{y}_k \leftarrow$ Normalized the coordinates by Equation(14)
  $\mathbf{pe}_k \leftarrow$ Compute the absolute position embedding by Equation(15);
  $\mathbf{X}_i^{pos} \leftarrow \mathbf{pe}_k$
end for
Pad the sequence $\mathbf{X}_i^{pos}$ with zeros to a fixed length L.
$f_i^S \leftarrow \mathrm{Proj}(\mathrm{Transformer}(\mathbf{X}_\mathrm{i}^\mathrm{S} + \mathbf{X}_\mathrm{i}^\mathrm{pos}))$
return f_i^S.

For the final sequence $\mathbf{X}^S \in \mathbb{R}^{L\times 17}$, where L denotes the length of the sequence, we insert an additional class vector to facilitate the similarity calculation in the contrastive learning. We assign a value of $+1$ to the category vectors for clips corresponding to the hotspot, while for all other cases, we assign a value of -1. Ultimately, l polygon vectors, along with their corresponding category vectors, collectively constitute the sequence $\mathbf{X}^S$. The complete Layout2Seq pipeline is detailed in Algorithm 2.

D. Absolute Position Embedding

Our proposed Layout2Seq incorporates the geometric properties of polygons but does not explicitly encode the topological relationships among them. Polygon-based topology is particularly relevant in the definition of hotspots. To capture positional relationships in serialized layout data, the Transformer architecture employs a position encoder that characterizes dependencies between tokens. However, existing methods [12], [14] commonly adopt a default position encoding strategy that represents polygon order without explicitly considering their actual spatial distances or topological relationships.

Algorithm 4 Inference process for our proposed GRC-HD

Require: A layout clip image $\mathbf{X}^I$, a predefined library of geometric operations $\{op(\cdot)\}$, $\mathcal{I}$ and $\mathcal{S}$, as returned by training process.

Ensure: the prediction result Y.

Assuming $\hat{Y} = HS$, compute the $\mathbf{X}_{hs}^S$ by Algorithm 2;
Assuming $\hat{Y} = NHS$, compute the $\mathbf{X}_{nhs}^S$ by Algorithm 2;
$Y \leftarrow$ *HardVoting*($\mathbf{X}^I$,$\mathbf{X}_{hs}^S$, $\mathbf{X}_{nhs}^S$) or
  SoftVoting($\mathbf{X}^I$, $\mathbf{X}_{hs}^S$,$\mathbf{X}_{nhs}^S$);
return Y.

HardVoting($\mathbf{X}^I$,$\mathbf{X}_{hs}^S$, $\mathbf{X}_{nhs}^S$):
$NUM_{HS} = 0, NUM_{NHS} = 0$
for $i = 1$ to $len(\{op(\cdot)\})$ **do**
  ${\mathbf{X}^I}_{tmp} \leftarrow op_i(\mathbf{X}^I)$
  $f^I = \mathrm{Proj}(\mathcal{I}({\mathbf{X}^I}_{tmp}))$;
  $f_{hs}^S, f_{nhs}^S = \mathrm{Proj}(\mathcal{S}(\{\mathbf{X}_{hs}^S, \mathbf{X}_{nhs}^S\}))$;
  $score_{hs}, score_{nhs} = \mathrm{Softmax}(\mathrm{sim}(f^I, (f_{hs}^S, f_{nhs}^S)))$;
  if $score_{hs} > score_{nhs}$ **then**
    $NUM_{HS} += 1$;
  else
    $NUM_{NHS} += 1$;
  end if
end for
return $\mathrm{argmax}(NUM_{HS}, NUM_{NHS})$

SoftVoting($\mathbf{X}^I$,$\mathbf{X}_{hs}^S$, $\mathbf{X}_{nhs}^S$):
$HS = \{\}, NHS = \{\}$
for $i = 1$ to $len(\{op(\cdot)\})$ **do**
  ${\mathbf{X}^I}_{tmp} \leftarrow op_i(\mathbf{X}^I)$
  $f^I = \mathrm{Proj}(\mathcal{I}({\mathbf{X}^I}_{tmp}))$;
  $f_{hs}^S, f_{nhs}^S = \mathrm{Proj}(\mathcal{S}(\{\mathbf{X}_{hs}^S, \mathbf{X}_{nhs}^S\}))$;
  $score_{hs}, score_{nhs} = \mathrm{Softmax}(\mathrm{sim}(f^I, (f_{hs}^S, f_{nhs}^S)))$;
  $HS \leftarrow score_{hs}$;
  $NHS \leftarrow score_{nhs}$;
end for
$HS \leftarrow \mathrm{Average}(HS), NHS \leftarrow \mathrm{Average}(NHS)$;
$score_{hs}, score_{nhs} = \mathrm{Softmax}(HS, NHS)$;
return $\mathrm{argmax}(score_{hs}, score_{nhs})$

In this article, we integrate spatial positional information of polygons into the sequential learning process of model $\mathcal{S}(\cdot)$ via the proposed APE. Specifically, for a given polygon P^k, the centroid coordinates (x_k, y_k) are computed from central moments as:

$$x_k = \frac{\mathbf{C}_{10}^k}{\mathbf{C}_{00}^k}, y_k = \frac{\mathbf{C}_{01}^k}{\mathbf{C}_{00}^k}. \tag{13}$$

To facilitate efficient training in the sequence model, we normalize coordinates by scaling them with respect to the dimensions of the input layout clip. Specifically, the normalized coordinates $\hat{x}_k$ and $\hat{y}_k$ are computed as:

$$\hat{x}_k = \frac{x_k}{W}, \hat{y}_k = \frac{y_k}{H}, \tag{14}$$

where W and H represent the width and height of the input layout clip, respectively. As shown in Fig. 5(a), we exclude the conventional position encoder typically used in transformer architectures due to the lack of continuous contextual

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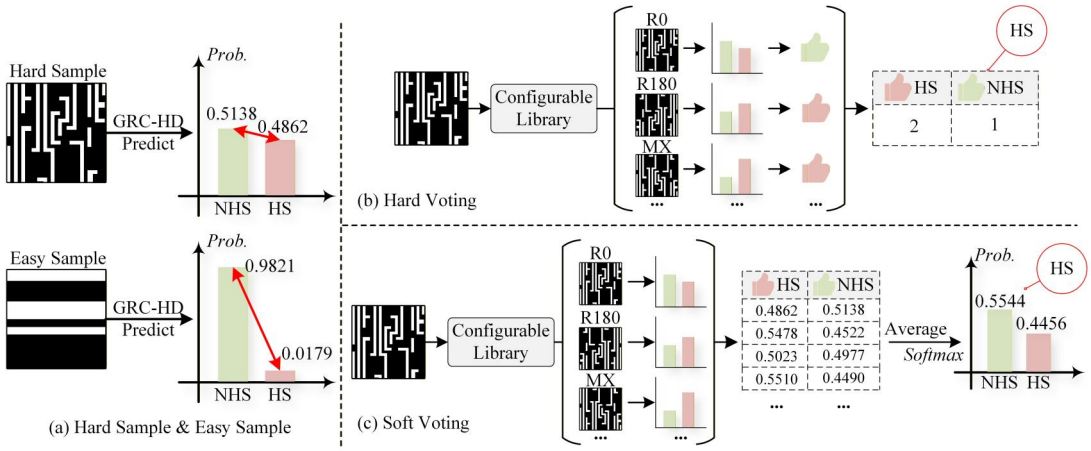

Fig. 6. Illustration of the definitions of hard and easy samples, along with the overall pipeline of hard voting and soft voting.

dependencies between polygons. Alternatively, we introduce a concise multi-layer perceptron (MLP) designed to encode the computed centroid coordinates, facilitating their direct integration into the sequence model. The absolute position embedding $\mathbf{pe}_k$ for the k-th polygon is defined as:

$$\mathbf{pe}_k = \sigma(\mathrm{FC}([\hat{x}_k, \hat{y}_k])), \tag{15}$$

and the resulting position-embedding sequence is $\mathbf{X}_i^{pos} = \{\mathbf{pe}_k\}_{k=1}^{l}$, where l denotes the number of polygons.

As shown in Fig 5(b), to model both geometric properties and topological relationships, we use the Transformer structure with the proposed APE. Consistent with other approaches utilizing position embedding, we adopt summation to fuse position information into the sequence representation. The modeling process for the i-th layout is detailed in Algorithm 3.

E. Inference and Self-Calibration Voting

During inference, supervised learning-based models classify test samples as hotspots or non-hotspots based on a learned decision boundary. However, constructing an effective decision boundary is difficult because hotspot detection datasets are typically class-imbalanced and exhibit low inter-class separability [13], [29]. In contrast, the proposed GRC-HD leverages contrastive learning to align image embeddings with corresponding sequence embeddings. Consequently, GRC-HD predicts a test sample by computing its similarity to class-specific sequence embeddings representing hotspot and non-hotspot categories. Specifically, as shown in Fig. 2(b), given a layout image $\mathbf{X}^I$, Layout2Seq generates two class-specific sequence embeddings, the hotspot sequence $\mathbf{X}_{hs}^S$ and the non-hotspot sequence $\mathbf{X}_{nhs}^S$, based on the input test sample. The image and its associated sequences are independently processed through the image encoder and the sequence encoder to obtain latent representations $(f^I, (f_{hs}^S, f_{nhs}^S))$. We then compute the similarity between the image embedding and each of the sequence embeddings, followed by a softmax function to produce the final classification probabilities.

In Section III-B, the proposed GRC-HD employs geometric-robust contrastive learning to enforce consistency in the image embedding space across geometrically transformed layout samples. Thus, the similarity between the resulting image embeddings and their corresponding sequence embeddings should remain invariant under these transformations. However, as shown in Fig 6(a), some hard samples exhibit ambiguous discriminability, in that the image embedding is similarly aligned with both HS and NHS sequence embeddings.

To enhance prediction robustness and reliability, this article proposes a self-calibration voting mechanism that combines test-time augmentation with ensemble inference, aggregating predictions from multiple transformed views of each sample. Specifically, the self-calibration process includes two alternative strategies: hard voting and soft voting. As shown in Fig 6, we independently apply the proposed configurable transformation library to each input image, consistent with the training pipeline, thereby generating distinct voters. Each voter is encoded by the image encoder and computes the similarity between the embedding and the NHS and HS sequence embeddings. As illustrated in Fig 6, for hard voting, each voter selects the class (NHS or HS) with the higher similarity score, and the final prediction is the majority label across voters. In contrast, soft voting aggregates the similarity scores across voters by averaging them and then applying softmax to obtain the final class probability distribution. The inference process with self-calibration voting is detailed in Algorithm 4. A comparative analysis of both voting strategies is presented in the experimental section.

TABLE I
STATISTICS OF THE DATASET

Benchmark	Training Set		Testing Set		Node(nm)
	#HS	#NHS	#HS	#NHS	
ICCAD-2012	1204	17096	2524	13503	28/32
ICCAD-2016	1300	9935	1301	2484	7
ICCAD-2019-1	467	17758	1001	14621	28
ICCAD-2019-2	467	17758	64310	65523	28

IV. EXPERIMENTAL RESULTS

All experiments were conducted using the PyTorch framework on a high-performance computing platform comprising

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AMD EPYC 9554 64-core CPUs and eight NVIDIA RTX A6000 GPUs with 48 GB of memory each, providing the computational throughput required for efficient model training and evaluation.

A. Dataset

The performance of the proposed GRC-HD method is evaluated on three standard benchmarks:

- *ICCAD-2012* [30]: This benchmark combines four layout datasets based on a 28 nm process with four metal layers, and one dataset based on a 32 nm process with a single metal layer. Following the data preprocessing described in [31], each dataset is divided into training and testing subsets. Each layout clip has a fixed size of $4.8\mu m \times 4.8\mu m$. These clips include both hotspot and non-hotspot layout patterns. Specifically, within each clip labeled as a hotspot, a central region with $1.2\mu m \times 1.2\mu m$ is defined as the hotspot area.
- *ICCAD-2019* [29]: This benchmark extends ICCAD-2012 with a more diverse and challenging set of hotspot patterns. ICCAD-2012 contains relatively few truly unseen patterns, which limits its effectiveness for evaluating the generalization of deep learning-based hotspot detection models. In contrast, ICCAD-2019 introduces a broader range of previously unseen hotspot patterns, enabling a more rigorous evaluation on novel and complex scenarios.
- *ICCAD-2016* [32]: The dataset includes four design layouts that have been scaled to match with extreme ultraviolet (EUV) lithography metal layer design rules. ASML's Tachyon Layout Manufacturability Checker (LMC) is utilized to simulate and analyze lithographic hotspots, with the resulting hotspot location labeled into the layout file. The simulation focuses primarily on evaluating critical lithographic process variations that significantly contribute to circuit reliability concerns, particularly edge placement errors (EPE), line-width necking, and pattern bridging. Following the [3], we randomly extract non-hotspot samples from the training region. We further apply data augmentation via random cropping of the original images, which allows a single cropped clip to contain multiple hotspot patterns and thereby increases detection difficulty.

Comprehensive dataset details are provided in TABLE I.

B. Implementation Details

The image encoder in GRC-HD leverages ViT with a patch size of 32, a latent embedding dimension of 768, and 12 Transformer layers. For sequence encoding, we adopt a BERT-based model with 12 Transformer layers, a hidden size of 768, and 12 self-attention heads. Since layout clips differ substantially from natural images and Layout2Seq relies on a layout-aware serialization rather than standard tokenization, we do not initialize either encoder from pre-trained backbone. Instead, both encoders are trained end-to-end from scratch. To ensure a fair comparison with existing state-of-the-art approaches, all experiments are conducted under consistent training settings.

TABLE II
COMPARISON ON ICCAD-2012 BENCHMARK WITH THE STATE-OF-THE-ART METHODS.

Method	Acc(%)	FA(%)	Time(s)
TODAES'2019 [35]	97.60	5.60	-
DAC'2021 [10]	98.25	6.10	-
TODAES'2022 [36]	98.90	9.90	30.0
TCAD'2022 [24]	98.77	18.50	**3.0**
DATE'2022 [16]	98.42	12.80	3.2
DATE'2023 [11]	96.20	6.40	7.5
GLSVLSI'2024 [22]	88.30	**0.20**	103.4
DAC'2024 [12]	98.49	8.30	9.9
TCAD'2025 [34]	99.78	5.29	-
TODAES'2025 [37]	98.10	4.00	6.5
CLI-HD [38]	99.72	1.20	10.0
Ours	**99.84**	0.54	20.2

TABLE III
COMPARISON ON ICCAD-2019 BENCHMARK WITH THE STATE-OF-THE-ART METHODS.

Methods	ICCAD-2019-1		ICCAD-2019-2		Average	
	Acc(%)	FA(%)	Acc(%)	FA(%)	Acc(%)	FA(%)
TCAD'2020 [39]	80.90	2.50	89.80	83.90	85.30	43.20
TODAES'2022 [36]	87.20	9.70	90.30	84.10	88.75	46.90
DATE'2023 [11]	91.60	8.60	90.50	83.90	91.05	46.25
GLSVLSI'2024 [22]	62.30	3.70	84.30	64.20	73.30	33.95
TCAD'2025 [23]	91.30	8.40	94.60	81.80	92.95	45.10
CLI-HD [38]	96.10	**1.00**	87.02	20.53	91.56	10.77
Ours	**97.90**	1.01	**90.61**	**15.32**	**94.26**	**8.17**

Specifically, the learning rate is set to 2×10^{-4}, with a batch size of 32 and a total of 100 training epochs. The Adam optimizer is employed with a weight decay of 1×10^{-2} and a momentum coefficient of 0.9. All clip images in the dataset are preprocessed via cropping and resizing to a resolution of 512×512 pixels to reduce computational overhead [4], [33], [34].

C. Performance Comparisons with Existing Methods

1) Results on ICCAD-2012: To evaluate the effectiveness of the proposed approach, we present comparison results on the ICCAD-2012 dataset. As shown in TABLE II, our proposed method demonstrates superior accuracy compared to other state-of-the-art algorithms (97.60% of TODAES'2019 [35], 98.25% of DAC'2021 [10], 98.90% of TODAES'2022 [36], 98.77% of TCAD'2022 [24], 98.42% of DATE'2022 [16], 96.20% of DATE'2023 [11], GLSVLSI'2024 [22] of 88.30%, DAC'2024 [12] of 98.49%, TCAD'2025 [34] of 99.78%, TODAES'2025 [37] of 98.10% and DAC'2025 [38] of 99.72% vs. 99.84% of ours). Moreover, our method mitigates the excessive false positives observed in prior work, reducing the false-alarm rate to 0.54% on ICCAD-2012. Finally, the proposed method achieves competitive performance while maintaining a trade-off between the accuracy and speed.

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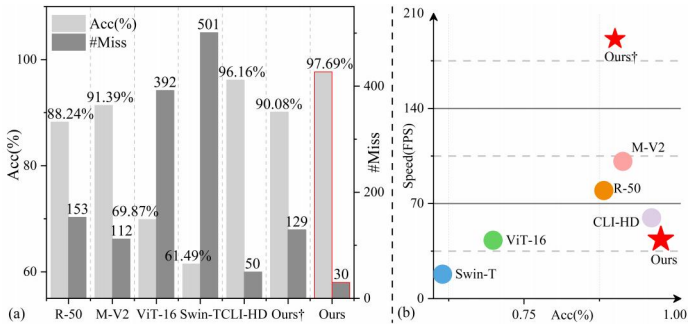

Fig. 7. Comparison with the deep learning-based methods on ICCAD-2016. (a) Detection accuracy compared with other methods. Bars with solid red borders indicate optimal results. † denotes our results of supervised fine-tuning. #Miss denotes the number of false negatives. (b) Detection speed compared with other methods.

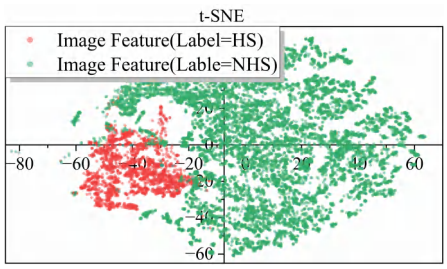

Fig. 9. Illustration of t-SNE result for the fine-tuning.

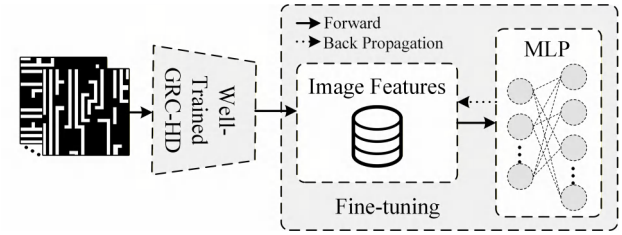

Fig. 8. Illustration of the pipeline for fine-tuning in GRC-HD.

TABLE IV
QUANTITATIVE COMPARISON OF SUPERVISED LEARNING AND CONTRASTIVE LEARNING.

	Acc(%)	FA(%)
Supervised Learning	79.23	46.91
Contrastive Learning	**99.84**	**0.54**

2) Results on ICCAD-2016: We evaluated four state-of-the-art backbone networks (ResNet50 (R-50) [40], MobileNetV2 (M-V2) [41], ViT-16 [42], and Swin-Transformer Tiny (Swin-T) [43]) using supervised learning on the ICCAD-2016 datasets, and compared them with our proposed method. As shown in Fig. 7(a), our method achieves an accuracy of 97.69% while reducing the number of missed detections to only 30. To evaluate generalization, we fine-tuned the learned representations with a supervised learning framework, denoted as Ours†. Fine-tuning the learned representations with two fully connected layers achieves 88.24% accuracy, comparable to R-50 (90.08%). Furthermore, as shown in Fig. 7(b), our fine-tuned model demonstrates significantly improved inference speed, achieving nearly 200 FPS and outperforming all comparative methods.

3) Results on ICCAD-2019: We perform comparative experiments on the ICCAD-2019 benchmark, which consists of two subsets, ICCAD-2019-1 and ICCAD-2019-2, both sharing a common training set. TABLE III reports the performance of our proposed method and six state-of-the-art hotspot detection algorithms, as presented in TCAD'2020 [39], TODAES'2022 [36], DATE'2023 [11], GLSVLSI'2024 [22], TCAD'2025 [23], and CLI-HD [38]. Our proposed framework achieves an average accuracy of 94.26%, outperforming other methods. Furthermore, GRC-HD enhances accuracy by an average of 2.7% and reduces the false alarm rate by 2.6% on the ICCAD-2019 dataset, compared to the previous version of CLI-HD [38].

D. Discussion of Fine-tuning for GRC-HD

In this experimental section, we propose a simple fine-tuning pipeline. As shown in Fig 8, we first extract image features using the well-trained GRC-HD model and store them for later tuning. Next, we apply supervised learning to train a classifier on these fixed features. To demonstrate the efficacy of the extracted features, we use a simple classifier consisting of only two fully connected layers. The qualitative results are presented in Fig 9, where we visualize the t-SNE [44] embeddings of the learned classification features. These embeddings clearly highlight the distinct separation between hotspot and non-hotspot feature distributions, demonstrating improved decision boundaries compared to prior supervised learning approaches. In addition, quantitative performance is provided in the Section IV-C2, which shows that the fine-tuned model achieves nearly a 4x speedup in inference while maintaining accuracy within an acceptable range of degradation.

E. Ablation Study

In the ablation studies, we evaluated the model's generalization under geometric variations by randomly applying geometric transformations to test samples at evaluation time. The resulting transformed datasets are denoted as ICCAD-2012++, ICCAD-2016++, and ICCAD-2019++, respectively. To validate the contribution of each proposed component, we use CLI-HD [38] without APE as the baseline.

1) Supervised learning vs. Contrastive learning: To evaluate the effectiveness of the contrastive learning framework proposed in this article, we compare it with the supervised-learning method on the ICCAD-2012++ dataset. For a fair comparison, the method adopts the same ViT backbone as our method, and we keep the data preprocessing pipeline identical to that used in GRC-HD. As shown in Fig 10, we present a qualitative analysis of the two learning methods. Fig. 10(a) shows the t-SNE projection of supervised features, where most HS features overlap with the NHS distribution, making them difficult to separate using a single decision boundary. In

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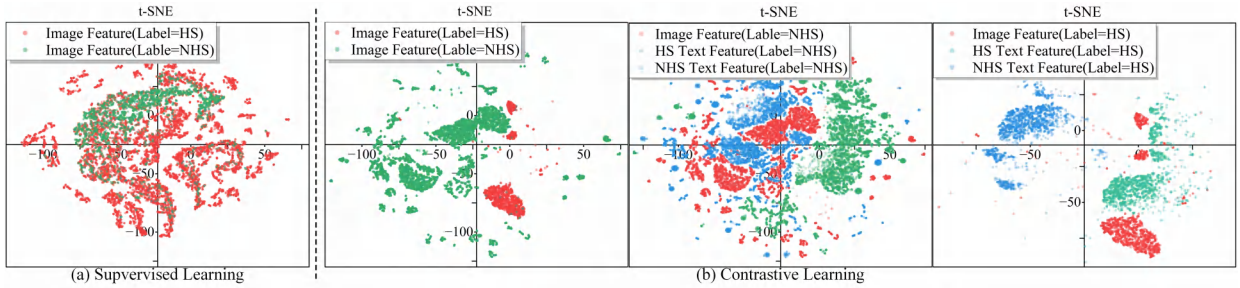

Fig. 10. Qualitative comparison of supervised learning and contrastive learning. (a) t-SNE visualization of the feature distribution from the last layer of the baseline model. (b) t-SNE visualization of the feature distribution from the last layer using the proposed GRC-HD.

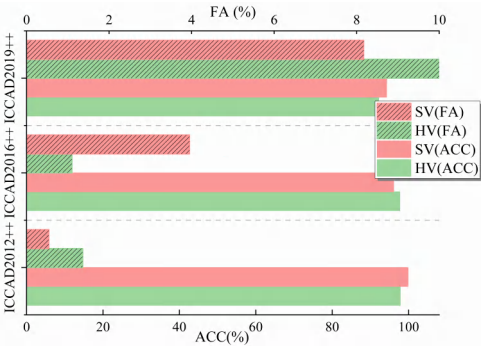

Fig. 11. Illustration of the effects of hard voting (HV) and soft voting (SV) in self-calibration voting.

TABLE V
QUANTITATIVE COMPARISON OF VARIOUS LAYOUT REPRESENTATION AND THE PROPOSED LAYOUT2SEQ ON ICCAD-2012.

	Acc(%)	FA(%)
Image-based [22]	88.30	**0.20**
Graph-based [16]	98.42	12.80
Language-based [12]	98.49	8.40
Layout2Seq-based	**99.84**	0.54

contrast, the left side of Fig. 10(b) shows the feature distribution produced by the GRC-HD, where contrastive learning promotes clearer separation between HS and NHS features.

Furthermore, for classification, contrastive learning relies on similarity calculations. The middle of Fig 10(b) presents the distribution of image and sequence features labeled as NHS, highlighting that NHS image features cluster closely with NHS sequence features. In contrast, the right of Fig 10(b) shows the distribution of image and sequence features labeled as HS, demonstrating that HS image features are closely aligned with their corresponding HS sequence features. The quantitative comparison results are presented in the TABLE IV, where the contrastive learning shows a significant advantage over the supervised learning, leading to a reduction in false alarm rate and an improvement in accuracy.

2) Effectiveness of proposed Layout2Seq: To validate the effectiveness of the Layout2Seq, we conduct a comparative evaluation of different representation methods for the hotspot detection. As shown in TABLE V, the image-based representation yields the lowest accuracy, despite achieving the fewest false alarms. In comparison, the graph-based representation with a graph neural network and the language-based representation with the Transformer show suboptimal performance relative to the proposed methods.

TABLE VI
ABLATION STUDY OF STRUCTURE FOR GRC-HD.

+APE	+GRT	+SCV	ICCAD-2012++		ICCAD-2016++		ICCAD-2019++	
			Acc (%)	FA (%)	Acc (%)	FA (%)	Acc (%)	FA (%)
-	-	-	87.00	13.78	90.54	13.97	90.21	20.14
✓	-	-	90.86	7.68	92.08	11.19	91.41	19.78
✓	✓	-	95.97	1.07	95.47	1.61	92.12	10.81
✓	✓	✓	**99.84**	**0.54**	**97.69**	**1.10**	**94.26**	**8.17**

3) Effectiveness of proposed APE: As presented in TABLE VI, we first evaluate the effectiveness of APE. Incorporating APE into the baseline model consistently improves accuracy while reducing false alarms across all datasets. Specifically, the accuracy for ICCAD-2012++ increases from 87.00% to 90.86%, while the false alarm rate decreases from 13.78% to 7.68%. For ICCAD-2016++, accuracy improves from 90.54% to 92.08%, and false alarms decrease from 13.97% to 11.19%. On ICCAD-2019++, the accuracy increases from 90.21% to 91.41%, with the false alarm rate reducing from 20.14% to 19.78%.

4) Effectiveness of proposed Geometric-Robust Training: The incorporation of Geometric-Robust Training (GRT) into the GRC-HD framework led to consistent improvements across all evaluated metrics. When combined with the APE, GRT further enhanced the model's accuracy. On the ICCAD-2012++ dataset, accuracy increased from 90.86% to 95.97%. Similarly, for ICCAD-2016++, accuracy improved from 92.08% to 95.47%, and for ICCAD-2019++, from 91.41% to 92.12%. Additionally, the implementation of GRT substantially reduced the false alarm rate by 6.61% on ICCAD-2012++, 9.58% on ICCAD-2016++, and 8.98% on ICCAD-2019++.

5) Effectiveness of proposed Self-Calibration Voting: As shown in TABLE VI, integrating Self-Calibration Voting (SCV) significantly reduces the false-alarm rate and improves accuracy across all three datasets. Section III-E introduces two SCV variants, namely hard voting (HV) and soft voting (SV). As shown in Fig. 11, SV achieves higher accuracy and lower false-alarm rate than HV on ICCAD-2012++ and ICCAD-

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TABLE VII
ABLATION STUDY OF THE CONFIGURABLE GEOMETRIC TRANSFORMATION LIBRARY.

+R90/R270	+R180	+MX	+MY	ICCAD-2012++		ICCAD-2016++		ICCAD-2019++	
				Acc (%)	FA (%)	Acc (%)	FA (%)	Acc (%)	FA (%)
-	-	✓	✓	99.08	0.64	98.00	2.25	93.53	9.62
-	✓	✓	✓	99.44	0.61	97.23	1.13	94.38	9.11
✓	✓	✓	✓	99.84	0.54	97.69	1.10	94.26	8.17

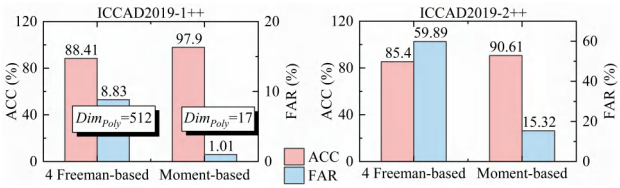
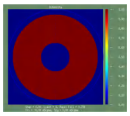
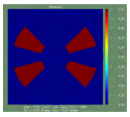


Fig. 12. Ablation on polygon shape descriptors for Layout2Seq on ICCAD-2019++ datasets.

TABLE VIII
L2 VARIATION ($\times 10^{-3}$) OF ICCAD-2016 LAYOUT PATTERNS UNDER DIFFERENT GEOMETRIC TRANSFORMATIONS, OBTAINED FROM LITHOGRAPHY SIMULATIONS WITH TWO ILLUMINATION SOURCES.

Source	Case	R90	R180	R270	MY	MX
	Testcase2	0.234	0.219	0.215	0.134	0.217
	Testcase3	0.979	1.081	1.001	1.024	0.830
	Testcase4	1.513	1.351	1.072	1.122	0.975
	Testcase2	**5.447**	0.080	**5.502**	0.084	0.073
	Testcase3	**10.64**	1.090	**10.57**	1.050	0.945
	Testcase4	**10.88**	1.214	**10.21**	0.991	1.029

2019++. In contrast, HV provides superior performance on ICCAD-2016++.

6) Impact of Different Shape Descriptors: In this section, we conduct an ablation study by replacing the moment-based polygon representation in Layout2Seq with a representative chain-code descriptor, namely the 4-Freeman boundary encoding. Specifically, for each polygon, we extract its 4-direction Freeman chain code along the boundary and convert it into a fixed-length embedding (Dim_{Poly}=512), which serves as the polygon sequence. Fig. 12 reports the results on two ICCAD-2019++ datasets. The moment-based descriptor used in Layout2Seq achieves higher accuracy and a lower false alarm rate than the 4-Freeman boundary encoding, while requiring a lower-dimensional representation. This performance gap likely arises because chain-code representations are sensitive to boundary discretization, vertex ordering, and sampling density, whereas moment-based statistics provide a compact and robust summary of polygon geometry.

7) Configurable Geometric Transformations under Different Illumination Sources: In advanced lithography, process effects are often anisotropic due to illumination asymmetry, scanner slit direction, lens aberrations, and polarization. As a result, common geometric augmentations (e.g., rotation and mirroring) may not be printing-equivalent under realistic process conditions. To quantify this effect, we performed lithography simulations on the ICCAD-2016 dataset under two representative illumination sources: (i) a symmetric annular source and (ii) an asymmetric quadrupole/4-pole (Quasar-like) source. TABLE VIII reports the L_2 variation ($\times 10^{-3}$) between the printed patterns of the original layout (R0) and those of geometrically transformed layouts (R90/R180/R270 and MX/Y). Under the annular source, the printed patterns remain highly consistent across all transformations, indicating that rotation/mirror operations are compatible with symmetric process conditions. In contrast, under the Quasar-like source, R90/270 lead to substantial deviations, whereas R180 and MX/Y remain comparatively stable. These results show that asymmetric illumination can break the equivalence of R90/270, while preserving a higher degree of consistency for R180 and MX/Y. Motivated by this observation, we design a configurable geometric transformation library in which the enabled operations are selected according to the assumed process/illumination conditions. We further conduct an ablation study to quantify the impact of different transformation subsets on hotspot detection performance. As shown in TABLE VII, these results support a geometric-robust design and demonstrate that the proposed transformation library can be adapted to different process conditions without changing the overall learning and voting framework.

F. Full-Layout Evaluation and Practical Workflow

1) Evaluation Points for Full-Layout Inference: To evaluate the practical applicability of the proposed hotspot detector beyond the benchmark setting, we construct an end-to-end workflow on three full-layouts from ICCAD-2016. As shown in Fig. 13(a), the workflow consists of four stages: 1) evaluation point selection, 2) hotspot detection via GRC-HD on layout clips extracted at these evaluation points, 3) post-pattern clustering of results to remove redundancy, and 4) analysis and inclusion of representative patterns into a library for subsequent reuse.

We investigate three evaluation points selection strategies: random sampling (ICCAD-2016), exhaustive traversal over the layout (ICCAD-2016*), and an EDA-assisted layout reduction to obtain representative evaluation points (ICCAD-2016†). Fig. 13(c) shows an example (Testcase3) where traversal generates dense candidates, while layout reduction reduces redundant regions. Fig. 13(b) reports the number of evaluated clips (#Total Samples), false alarms (#False Alarm), clusters

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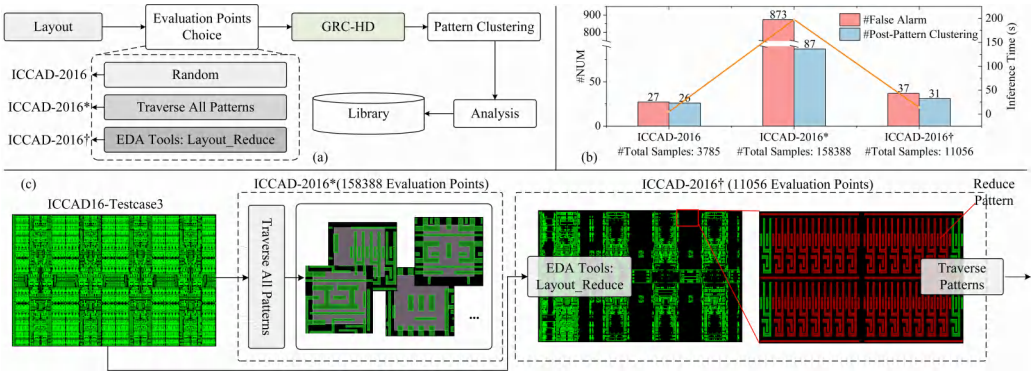

Fig. 13. Evaluation workflow and evaluation points selection on ICCAD-2016. (a) Practical deployment flow. (b) Comparison of three evaluation points selection strategies. (c) Visualization on ICCAD-16-Testcase3 illustrating dense candidates from traversal and the reduced representative regions.

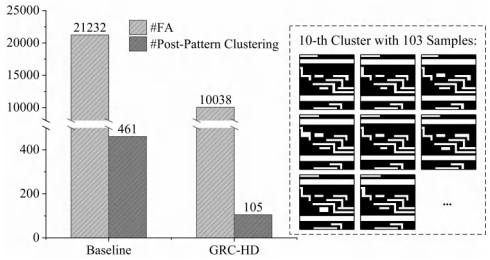

Fig. 14. Post-pattern clustering of false positive on ICCAD-2019-2++.

after post-pattern clustering (#Post-Pattern Clustering), and the overall inference time. We adopt DBSCAN [45] for post-clustering. Random sampling evaluates 3785 clips and produces 27 false alarms (26 clusters). Exhaustive traversal evaluates 158388 clips and yields 873 false alarms, which are compressed to 87 clusters after post-pattern clustering. With EDA-assisted layout reduction, only 11056 clips are evaluated, resulting in 37 false alarms (31 clusters), while substantially reducing the inference time compared to exhaustive traversal. These experiments highlight that GRC-HD can be integrated into practical full-layout workflows under different evaluation point generation policies.

2) Clustering of False Positive Samples on ICCAD-2019-2++: In practical deployment, the absolute number of false alarms may be large due to the massive number of candidate patterns in full-chip layouts. Nevertheless, false alarms are often highly redundant and can be reduced to a small set of representative patterns. Thus, we collect all false positive samples on ICCAD-2019-2++ and perform DBSCAN for pattern clustering. Fig. 14 reports the total number of false alarms (#FA) and the number of clusters after post-pattern clustering (#Post-Pattern Clustering). The baseline (CLI-HD) produces 21232 false alarms, which are reduced to 461 clusters. In contrast, GRC-HD yields 10038 false alarms and only 105 clusters, indicating that most false alarms can be summarized by a limited set of representative patterns. We further visualize representative samples from one cluster (the 10-th cluster containing 103 samples). These results show that post-pattern clustering substantially compresses redundant false alarms, facilitating practical review and library construction.

3) Application to Pattern Clustering: Beyond hotspot detection, the learned GRC-HD embedding can be directly reused for pattern clustering, which is commonly employed for redundancy reduction and library construction in DFM flows. TABLE IX compares our clustering results with two prior works (DAC'2017 [46] and TODAES'2023 [47]) on ICCAD-2016 Testcase3 and Testcase4, reporting #Cluster, maximum cluster size ($C_{\max}$), and runtime. We also use a gpu-accelerated DBSCAN for clustering. For Testcase3, all methods obtain 8 clusters, while our method runs fastest (0.024s on GPU vs. 2.00s and 0.091s on CPU). On the larger Testcase4, our method completes clustering in 2.88s on GPU, significantly faster than 198s and 10.3s on CPU. Although the cluster counts differ slightly, $C_{\max}$ is comparable, indicating similar redundancy compression. These results suggest that GRC-HD embeddings provide a scalable feature space for large-scale pattern grouping beyond hotspot detection.

V. Conclusion

In this work, we propose GRC-HD, a hotspot detector that leverages layout topological representation and geometric-robust contrastive learning with self-calibration voting. We develop a contrastive learning that aligns feature representations across binary images and layout sequences. Furthermore, we employ a Siamese training with a configurable geometric transformation library, and we use the same library during inference to aggregate predictions. Layout2Seq encodes polygon geometry and absolute positions to better capture the layout's topological structure. Compared with our previous framework, GRC-HD improves accuracy by up to 12.84% and reduces false alarms by up to 13.24% under transformed evaluation, demonstrating that geometric-robust training and self-calibration voting substantially enhance robustness. Moreover, the GRC-HD surpasses previous methods, achieving a maximum accuracy improvement of 6.3% and reducing the false alarm rate by up to 25.78%. Future work will explore the application of topological and contrastive learning techniques across diverse downstream tasks beyond hotspot detection.

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TABLE IX
PATTERN CLUSTERING RESULTS ON ICCAD-2016 TESTCASE3 AND TESTCASE4.

Dataset	#Evaluation Points	DAC'2017 [46]			TODAES'2023 [47]			Ours		
		#Cluster	$\#C_{max}$	Time(CPU)	#Cluster	$\#C_{max}$	Time(CPU)	#Cluster	$\#C_{max}$	Time(GPU)
Testcase3	5068	8	3335	2.00s	8	2816	0.091s	8	3968	0.024s
Testcase4	264824	17	198340	198s	16	198340	10.3s	19	202630	2.88s

ber: KYCX25_0394) and the National Natural Science Foundation of China (62341408). During the preparation of this work, the authors used ChatGPT solely to improve language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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